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20 years of monitoring CO₂-injection at Sleipner

Anne-Kari Furre

a*

, Ola Eiken

b

, Håvard Alnes

a

, Jonas Nesland Vevatne

a

, Anders Fredrik

Kiær

a

aStatoil ASA, Forusbeen 50, N-4035 Stavanger, Norway
bQuad Geometrics, Kjøpmannsgata 5, 7013 Trondheim, Norway

Abstract

The Sleipner CO₂ injection project was the world's first industrial offshore CO₂ Capture and Storage (CCS) project with more than 16 Mt CO₂ injected since 1996. Key monitoring insights from Sleipner are the dual interpretation of seismic and gravimetric monitoring surveys to quantify the free CO₂ mass changes and plume geometry development as a function of time. The learnings from Sleipner have contributed to making guidelines for monitoring future CCS injection projects, showing that selection of monitoring technology and the timing and extent of monitoring surveys should be case specific and risk based, while also taking into account the long term nature of CCS projects.

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1. Introduction

The Sleipner Vest gas field (Figure 1) was discovered in 1974, and put on stream in 1996, in a combined development with the Sleipner Øst condensate gas field. One challenge was the high CO₂ content (approximately 9 %). To meet the sales gas requirements, avoid pollution to the atmosphere and in response to the Norwegian CO₂ emissions tax, Statoil and partners decided to capture and inject the CO₂ into the saline, highly porous Utsira Fm. aquifer near the Sleipner Øst production platform [1], [2], [3]. CO₂ injection commenced through injection well 15/9-A-16 on September 15th1996. During the first year of injection

there were challenges related to injectivity, caused by sand influx, but these challenges were remediated by a re-perforation and installation of a gravel pack in August 1997 [3]. Since then injection has been stable, with a yearly rate of approximately 0.9 Mt during the first

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years, reducing slightly during later years, due to reduced gas flow from Sleipner Vest. Since 2014, CO₂ from the

Gudrun gas field has also been processed via the Sleipner CCS facility.

Figure 1: Left: location map (courtesy of Uwe Dederling). Middle: the Sleipner fields with the outline of the CO₂ plume in 2013 indicated (base map courtesy of NPD). Right: A seismic difference section illustrating the extent of the CO₂ plume in 2013.

An extensive geophysical and environmental monitoring programme has been deployed, as further described

below. None of the monitoring data have indicated any CO₂ release from the storage unit. As there were no

monitoring guidelines or regulations in place at the start of the Sleipner injection, Statoil and partners chose to test a

large range of methods, with frequent repeats and dense coverage. Suggestions for drilling a dedicated monitoring

well were rejected early on in the project on the grounds of cost and added risks, and the project has successfully

focused on use of remote geophysical monitoring methods.

In this paper we will focus on the lessons learned from monitoring during these 20 years, with particular emphasis

on seismic monitoring and the latest gravimetric results. We will examine the specific learnings and its implications

for reservoir understanding, but we will also discuss how the experiences from Sleipner can be employed when

planning new CCS projects.

Nomenclature

Mt million metric tons

2. CO₂ storage monitoring

Sleipner CO₂ injection is conducted under the Norwegian Petroleum Law, and CO₂ storage and monitoring was first included explicitly in a regulation announced December 8 2014 and since then a formal permit is in place. The wording in the Norwegian law is consistent with the EU directive 2009/31/EC. The monitoring objectives fall into three main categories:

- x Conformance monitoring: ensuring that the behaviour of CO₂ in the reservoir is understood.
- x Containment monitoring: ensuring that CO₂ stays within the storage unit.
- x Contingency monitoring: assessing effect of contingency measures in the case of leakage.

The regulations specify detailed directions for monitoring the facilities and wells. For the wider storage complex there are no detailed requirements, giving an operator a degree of freedom in planning the monitoring programme.

We will in this paper focus on our experiences with conformance and containment monitoring, and in particular on remote geophysical monitoring. The best methods available both for containment and conformance monitoring are typically remote geophysical techniques. The experience from Sleipner mainly comes from towed seismic streamers and gravimetric surveys, but we will also discuss the potential of other remote sensing technologies. Seismic sensors and (where applicable) sources could be towed in the water, or placed on the ground, at seabed, or in wells. They could be retrievable or permanent. Both seismic and gravimetric methods have been used for monitoring oil and/or gas fields [4], [5], in addition to CO₂ injection sites [6] [7]. Repeated resistivity measurements in wells are also an established technology [8], [9]. Recent research indicates that under favourable conditions, CSEM can be used for monitoring CO₂ [10], [11], but CSEM has not yet been applied in the time-lapse sense, neither for monitoring CO₂, oil or gas fields. In an offshore setting, seabed and well operations are relatively expensive compared to surface operations and a case specific cost benefit analysis should be conducted to arrive at the best solution for a new CCS project. At Sleipner a total of ten 3D seismic surveys and four gravity surveys have been acquired to date, and these have contributed to a detailed understanding of CO₂ behaviour in the storage unit. We discuss these in further detail below.

Wellhead pressure, temperature and rate have remained stable from production start-up, but due to the fact that the CO₂ at the wellhead is at the liquid/gas phase boundary, it is not possible to relate wellhead pressure to bottom-hole pressure without knowing the gas/liquid ratio. Down-hole pressure gauges would have been valuable, but the technology was not readily available in 1996. A Controlled Source Electromagnetic (CSEM) test line was acquired over the field in 2006, but the resolution was found inferior at the time. As it was challenging enough to detect the CO₂ plume, this line was consequently not repeated. The seabed has been surveyed with high-resolution acoustic imaging and photo mosaic, to investigate the potential of escape release structures. Finally chemical sampling of sediments and water column has been conducted, to search for potential increased CO₂ levels. None of the above monitoring techniques have indicated any leakage from the Sleipner CO₂ injection site.

3. Seismic - ideal for containment monitoring

When entering the reservoir, CO₂ is normally in supercritical state. This provides a strong sonic velocity contrast to the initially brine filled reservoir and favourable conditions for seismic monitoring. The seismic programme at Sleipner consists of repeated 3D towed seismic surveys, acquired in 1999, 2001, 2002, 2004, 2006, 2008, 2010, 2013, and 2016, respectively. The 2016 dataset was not available for analysis at the time of writing. These repeated surveys form, together with a base survey from 1994 (prior to injection start) a unique dataset which images the plume development [12, 13]. The early surveys were mostly conducted as research projects, while during later years the asset owner has been in charge of the monitoring. Seismic acquisition technology has improved significantly over the last two decades. Most surveys were acquired using conventional streamers, with increasing number of receiver cables over time. Also source and receiver configurations, and even tow direction have varied over the years. The six first acquisitions were combined with other seismic acquisitions in the area to save cost, and tow depths were not optimized for the CO₂ monitoring at shallow depths, neither in the initial base line nor the later repeats. Two surveys were acquired using broadband technologies; dual source in 2010, and slanted cable in 2013. The repeated datasets have all been through a common

time-lapse processing to enhance repeatability. Consequently, the image quality of the broadband surveys might be suboptimal, because all datasets were time-lapse processed to match the quality of the base survey. Despite these challenges related to variation in acquisition parameters and image quality, all surveys have been valuable for understanding the CO₂ plume development. This favourable outcome is attributed to the time-lapse processing and the large contrast in acoustic properties between the in-situ saline aquifer and the injected CO₂. The seismic response arising from this large acoustic impedance contrast is illustrated in Figure 2, which shows a vertical seismic section along the crest of the gentle anticline below which CO₂ is injected. Already at the time of the first time-lapse repeat survey (in 1999) nine bright reflections were identified. These reflections are interpreted as interfaces between sand layers saturated with CO₂, and thin shale layers. The sand layers are between 20-30m thick with the shales being typically 1-1.5m thick [2]. Well data confirm that thin shale layers which could act as flow baffles exist internally in the reservoir, but with the exception of one thicker [c. 6m thick] shale near the top of the storage reservoir it is not possible to correlate these shale layers between wells [2], or to identify them in the baseline seismic survey. CO₂ thus acts to illuminate the internal reservoir architecture, similar to the use of contrast dye in medical imaging. In the early years the CO₂ signatures in the shallower layers (6-9) were spatially small, and the deeper reflectors were easy to interpret. In more recent data, imaging is better for layers 5-9, whereas layers 1-4 are challenging to interpret (Figure 2). The reason for this degradation of the deeper signals is probably a combination of inelastic attenuation, transmission loss of signal through the CO₂ layers and CO₂ migration/dissemination [14].

Figure 2: Left: Map of the two-way time interpretation of the uppermost layer of the CO₂ plume, interpreted on the 2013 seismic time-lapse processed data. The yellow line corresponds to the position of the vertical sections in the middle and right vertical plots, representing the 1994 baseline and the difference between the 2013 and 1994 seismic data, respectively. The white line on the map represents the injection well, with perforation interval indicated as parallel white lines, whereas the magenta circle represents the nearest exploration well.

Up until approximately 2008, the temporal thicknesses of the CO₂ filled layers were below maximum seismic tuning, and layer thicknesses are nearly proportional to amplitude [12] [15]. Using an amplitude-

thickness

relationship, layer thicknesses down to meter scale can be observed, however, temporal thickness

(separating top

and base) of such thin layers cannot be separated. For the conventional time-lapse seismic data, having a dominant

frequency of approximately 35 Hz, the temporal tuning thickness between top and base layer is approximately 10 ms

[16].

Several attempts have been made at increasing the seismic resolution, both through acquisition and data analysis.

A 2D seismic survey was acquired in 2006, using source and receiver tow depths of only 3 m, as opposed to 6 m and

8 m for most of the 3D surveys. This pushes the frequency notches beyond the spectrum of the reflected pulses from

the CO₂ target. Despite the fact that the 2006 2D data could not be 3D migrated, they proved the potential of better

resolving the CO₂ layering in the reservoir, having a peak frequency of approximately 50 Hz, and giving a temporal

tuning thickness of approximately 7 ms [17]. The 2010 dual streamer data was studied using image processing in

addition to the time-lapse processing, which permitted pre-stack wave field separation and removal of the receiver

ghosts. This enabled a retrieval of higher frequencies, with a peak frequency of 50 Hz, similar to the 2D dataset [12].

Recent developments in spectral decomposition analysis have further enhanced the lower resolution limit of the

seismic data, utilizing the broader frequency spectrum [17] [15].

While resolution is limited, another important limitation is the seismic wave sensitivity to compressional velocities. Velocities change significantly as the fluid changes from pure brine to a brine with low CO₂ saturation;

much less as CO₂ saturation increases further. Repeated seismic data is therefore an excellent containment

monitoring tool to identify small CO₂ saturations in the reservoir, and potential leakages, but less successful in

quantifying further CO₂ saturation increases. Seismic data are also an important tool for conformance monitoring,

although additional data (especially reservoir rock properties and in situ fluid properties) are also important for that purpose.

4. Gravity monitoring – contributing to conformance monitoring

In contrast to velocity, density is linearly related to saturation. High-precision gravity monitoring offers an

independent measurement of density changes, and consequently of saturation. In 2002 permanent seabed

benchmarks were installed over the CO₂ plume at Sleipner and a baseline for gravity monitoring was acquired the same year [7]. Subsequent surveys were conducted in 2005, 2009, and 2013 [18]. The outline of the benchmarks is shown in Figure 3. In the original outlay only benchmarks 1-30 were installed, based on the need for calibration points (stations 1-20) and the plume extent at the time (estimated from seismic data; stations 7-10, and 21-30). The western part of the calibration line was acquired to compensate for gravity effects caused by water influx into the deeper gas condensate reservoir at Sleipner Øst (outlined in green). The eastern part of the line acts as reference stations in an area undisturbed both by injection and production. As the CO₂ plume has expanded over time, new benchmark stations have been installed (stations 31-43), covering a larger area.

Figure 3: Left: The gravity survey layout, with contours of the CO₂ plume and the deeper Sleipner East Ty gas-condensate reservoir. Right: Inverted CO₂ thickness change based on gravity change from 2002 to 2013, along with the CO₂ outline from seismic in 2013. Red arrows show decrease in measured gravity, blue arrows increase. The signal from Ty Fm. has been removed along with other signals mentioned in the text.

Gravimeters need repeated measurements due to instrumental drift. This calibration was performed by frequent visits to the eastern benchmark stations of the calibration line. Further error reduction was achieved by making redundant measurements, both through repeated visits at each station, and by using several gravimeters. In addition to density changes in the Utsira Fm, the gravity measurements are influenced by several other factors, which will need to be accounted for in order to identify the pure CO₂ response [7], [18]. First, ocean tidal fluctuations are accounted for using pressure gauges both on the gravity survey tool and stationary reference pressure gauges on the seafloor. Second, scouring by sea-bottom currents has lowered some of the benchmarks more than 10 cm into the sediments over time. This was also compensated for using the stationary pressure gauges. In addition, to reduce this error in the future, the new 2013 benchmarks were formed as mats instead of the conical shape which was used previously. The final effect influencing the gravimetric signal is hydrocarbon production and water influx in the deeper condensate reservoir at Sleipner Øst (the Ty and Hugin Fms). This leads to a gravity field increase with higher amplitude and longer wavelength than for the CO₂ influx in the Utsira Fm.

All these effects were inverted for during data processing to obtain the true gravimetric response of CO₂ replacing water in the Utsira Fm. This response is in the order of tens of PGal, whereas gravity repeatability has improved from maximum 4.3 PGal in the earlier surveys to 1.1 PGal in 2013. Based on the gravimetric measurements, and the plume geometry from seismic data, a mean in situ CO₂ density of 720 ± 55 kg/m³ is found, neglecting the effects of dissolution of CO₂ into the formation water. The gravimetric measurements can also be used to estimate an upper bound for the CO₂ brine dissolution rate, by assuming that the temperature in the reservoir is known. Within measurement uncertainty this gives a dissolution rate between 0 % and 2.7 % per year. This adds an important constraint to laboratory estimates of CO₂ dissolution in the formation water [19], an important process which reduces the long-term risk of leakage, because brine with CO₂ dissolved is heavier than pure brine and will sink to the bottom of the formation. In conclusion, within the measurement uncertainty the current amount of supercritical CO₂ in Utsira Fm. detected from gravity monitoring is the same as the injected amount of CO₂.

5. Lessons learned

Containment monitoring of CO₂ at Sleipner has been demonstrated by a total of nine repeated seismic surveys. Both seismic and gravimetric data have contributed to conformance monitoring. An important spin-off from the project has been the wealth of knowledge that has evolved through monitoring, and which has been shared with the scientific community over the twenty years of operation. Statoil and the Sleipner Licence partners have released all the seismic, gravity, and CSEM data acquired up to and including 2009. These data are available upon request and have been used for a wide range of applications, such as improving reservoir characterization, constraining flow modelling, and developing new techniques for seismic inversion and spectral decomposition. More than 150 scientific papers, conference abstracts, and several PhD and Master theses have used data from the Sleipner CO₂ project datasets.

5.1. Improved reservoir understanding

The data from the Sleipner CO₂ project have been widely used as constraints for reservoir flow modelling since project inception. In 2011 a benchmark reservoir model was made available to the scientific community (

[20], [21]

and references therein). This first benchmark model covers only the uppermost layer, whereas subsequent reference models cover the two uppermost layers [20]. The reason for focusing model efforts on the upper layers is that only these layers could be interpreted on the baseline seismic survey, prior to injection start, thus forming the basis for surface grids in the reservoir model. Reservoir properties were provided as part of the models [20], and the released repeated seismic data from injection start until 2008 act as a reference for the simulation results. The time-lapse seismic data show that the upper CO₂ layers tend to follow topographic highs. After some years of accumulation underneath the topographic high above the injection point, the CO₂ has spread in a relatively narrow corridor, and started accumulating below a structural high approximately 3 km north/north-east of the injection point. In general, results from several modelling studies have found it difficult to match the details of the time-lapse seismic [22], [23], [20], [21], [14], [24], [25]. This is due partly to challenges understanding the underlying physics of the CO₂ flow, and partly due to uncertainties in geological assumptions. On the microscopic scale the CO₂ injection into brine filled sandstone in the Utsira Fm is a drainage process but imbibition can occur with plume migration. There is still some debate on whether the upscaled flow process is dominated by viscous or gravity forces, although increasingly a gravity-dominated interpretation is being proposed [26, 27]. Viscous dominated models will typically not be able to reproduce the northwards corridor observed in the later seismic surveys [23], [25], whereas gravity dominated models predict an affinity to travel too far north too early [21], [25]. Geological assumptions, such as the number of feeder channels up to the top layer, permeability, a correct representation of the top seal topography, and temperature distribution will all have a strong effect on the final flow pattern, and contain some uncertainty [25]. It is not within the scope of this paper to conclude on the way forward for flow simulations, but the Sleipner time-lapse seismic data have been crucial for generating new insights that are valuable for later CO₂ injection projects. Similarly, the excellent time-lapse dataset has spurred scientists to use it for developing and testing new inversion schemes, both inverting from seismic amplitudes to impedances, and further to rock physics properties. The data

have been used for full, partial, and pre-stack 2D and 3D inversions, and in later years also Full Waveform Inversion (FWI), and joint inversion between seismic and gravimetric data [28], [29], [30], [31], [32], [33] [34], [35].

Seismic inversion studies can at their best lead to improved understanding of the reservoir and time-lapse observations because they provide a quantitative estimate of elastic properties. The effect of the seismic wavelet is removed, so that elastic property changes are confined to where they belong, and not smeared out by wavelet effects. Further, with a good rock physics model, inversion for saturation changes can also be conducted. However, the limitations in the available Sleipner data mean that only inversion for compressional velocity will give meaningful results. Inversion for shear velocity or density will be too error prone. Due to uncertainty in choice of saturation model, compressional velocity sensitivity to gas saturation is uncertain, meaning that without additional constraints, CO₂ saturation estimates are not conclusive from purely seismic inversion [31]. Because of this the Sleipner inversion studies have been mainly used for method development, but seismic inversion could be a complementary tool in CO₂ monitoring.

5.2. Optimising monitoring programmes

CO₂ monitoring regulations give the site operator considerable freedom, and consequently a large responsibility, when designing a monitoring programme to satisfy the overall requirements. The main factors to take into account are which technologies to use, and the spatial and temporal sampling intervals; factors that influence both cost and information content. We will discuss the specific learnings from Sleipner, and go on to discuss monitoring in more general terms.

The experience from Sleipner has shown that seismic monitoring is the most efficient containment monitoring tool, due to the ability to remotely cover a large area, the strong response expected even for small CO₂ saturations, and consequently the ability to detect small changes (of order one meter vertically and 5-10 meters laterally).

Seismic survey interval at Sleipner was quite short; two-three years. No big surprises have emerged from one survey to the next. This dense series of data points can be used to estimate the error introduced by dropping one or more surveys [36], and consequently to evaluate the overall cost of the monitoring programme compared to the gain of

retrieving more frequent data. The most recent survey at Sleipner was acquired four years after the previous one, reflecting confidence in previous measurements.

With the experience gained from such a well-behaved case as the Sleipner injection site we suggest that the monitoring strategy for CO₂ injection sites could be adjusted in future. For Sleipner we had fairly frequent seismic repeats for the first 15 years of injection (on average every second year), and then increased the time between surveys to four years. One approach for future monitoring projects could be to have the first repeat as soon as possible after injection start (depending on expected response perhaps after about one year), but to increase the survey interval if the injection goes according to plan. Such a programme should anyway be designed flexibly, so that if other monitoring devices indicate anomalies (e.g. anomalous well pressure behaviour or indications of leakage at the surface) new seismic surveys could be mobilized.

Industry experience demonstrates that seismic acquisition prices have varied considerably over time (for the Sleipner injection the seismic prices have varied by an order of magnitude during the project's lifetime). A robust monitoring project must therefore account for net present value and varying survey prices.

As discussed above, streamer seismic is a good compliance monitoring tool. Gravimetric measurements can provide a good supplement for shallower cases, because of their ability to detect mass changes. The main limitation of gravimetric monitoring is the spatial resolution, which is low compared to seismic resolution. Gravimetric measurements have continuously improved precision over time as the technology has matured.

The Sleipner CO₂ injection site has been a unique playground for monitoring, due to its long history with regular operations, and favourable geological conditions. The high-permeable and large reservoir at Sleipner may, however, be less representative of more heterogeneous, complex reservoirs where higher pressure build-up could cause a stronger stress response. During the Snøhvit CO₂ injection, pore pressure build-up was significant [37]. In such a case, the need for extracting pressure information from the remote monitoring becomes stronger.

Attempts have been made at separating pressure and saturation effects from the seismic data [38], [39]. It is possible to take remote CO₂ injection monitoring one step further, by installing sensors on the seabed instead of operating streamer surveys. The added value of three component data would provide separation of

compressional and shear waves, which could in turn provide aid in pressure and saturation separation. The ability to monitor geomechanical changes through shear-wave splitting is an important new use of multicomponent seismic

data [40] [41]. Retrievable Ocean Bottom Seismic (OBS) surveys are, however, usually several times more costly

than streamer surveys, meaning that retrievable systems are not realistic from a cost perspective.

Permanent

Reservoir Monitoring (PRM) seismic systems on the other hand could be an option. Though these have relatively

high initial installation costs, they might over time be competitive with towed streamer surveys, depending on

sampling frequency. A permanent seismic system in listening mode could detect potential microseismic activity

[42], and could act as a trigger to acquiring a repeated seismic survey. Permanent systems usually have superior

repeatability over retrievable systems, and could in passive mode be used for noise analysis. Several oil- and/or gas

fields in operation are monitored by such PRM systems today [43] [44], [45] [46]. A permanent layout could in

particular compete with towed streamer data if there are sea surface installations at the field. Such installations

would prevent the towed streamers from accessing the area near the installations, which would in turn require

expensive undershoots. A permanent system should also be planned in a flexible way, such that if the CO₂ moves

towards the border of the survey area, a new lay-out could be added on to the existing one.

6. Conclusions

The CO₂ injection project at Sleipner has been both a technical and economic success. Remote geophysical

monitoring has convincingly demonstrated that the CO₂ stays safely in the storage unit. Overall the monitoring

experience at Sleipner shows that:

- x CO₂ stays safely in the storage unit

- x Downhole pressure and temperature gauges are needed for obtaining good control of the injection in-situ

conditions and should be implemented for future CO₂ injection projects.

- x Repeated seismic surveys have been crucial for ensuring both containment and conformance monitoring.

Currently no good technology alternative exists.

- x By acquiring both gravimetric and seismic monitoring, it is possible to combine the free CO₂ mass change and

plume geometry data, and make an estimate of dissolution in formation water. This is important for

longer term

predictions. Together the datasets give much richer information than when handled separately.

Further learnings for future projects is that evaluations of specific monitoring design details (such as selection of

monitoring technology, timing and extent of monitoring surveys) should be case specific and risk based.

The

monitoring strategy should be flexible, providing the opportunity to combine planned and triggered repeats, and

taking into account the long term nature of CO₂ injection projects. The monitoring strategy should also be able to

adapt as technology improves, as it has done over the 20-year Sleipner monitoring history.

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